

Learning a Deep Multi-Scale Feature Ensemble and an Edge-Attention Guidance for Image Fusion

Jinyuan Liu[✉], Xin Fan[✉], *Senior Member, IEEE*, Ji Jiang, Risheng Liu[✉], *Member, IEEE*, and Zhongxuan Luo

Abstract—Image fusion integrates a series of images acquired from different sensors, *e.g.*, infrared and visible, outputting an image with richer information than either one. Traditional and recent deep-based methods have difficulties in preserving prominent structures and recovering vital textural details for practical applications. In this article, we propose a deep network for infrared and visible image fusion cascading a feature learning module with a fusion learning mechanism. Firstly, we apply a coarse-to-fine deep architecture to learn multi-scale features for multi-modal images, which enables discovering prominent common structures for later fusion operations. The proposed feature learning module requires no well-aligned image pairs for training. Compared with the existing learning-based methods, the proposed feature learning module can ensemble numerous examples from respective modals for training, increasing the ability of feature representation. Secondly, we design an edge-guided attention mechanism upon the multi-scale features to guide the fusion focusing on common structures, thus recovering details while attenuating noise. Moreover, we provide a new aligned infrared and visible image fusion dataset, RealStreet, collected in various practical scenarios for comprehensive evaluation. Extensive experiments on two benchmarks, TNO and RealStreet, demonstrate the superiority of the proposed method over the state-of-the-art in terms of both visual inspection and objective analysis on six evaluation metrics. We also conduct the experiments on the FLIR and NIR datasets, containing foggy weather and poor light conditions, to verify the generalization and robustness of the proposed method.

Index Terms—Image fusion, deep feature learning, attention mechanism.

I. INTRODUCTION

IMAGE fusion [1]–[4] plays a critical role in computer vision community since it can provide informative and reliable images for high-level vision tasks including object detection [5]–[8], pedestrian re-identification [9], [10], face recognition [11], semantic segmentation [14], [15], and other fields [16]–[19]. In particular, the combination of visible and infrared images has remarkable advantages. Commonly, visible images contain abundant textural details with high spatial resolution. However, visible images often present poor visual quality under darkness or harsh environmental conditions. Complementarily, discriminative thermal radiation captured by infrared sensors is able to highlight the thermal targets even in unfavorable conditions, *e.g.*, severe weather and poor lighting. The limitation of thermal images lies in vague textural details due to their low spatial resolution. Therefore, it is worthwhile to integrate the complementary strengths of infrared and visible images into a composite image favorable for human visual perception and computer vision applications, *e.g.*, supervisory control and autonomous driving.

The taxing point for infrared and visible fusion is how to extract typical characteristics from two sources and how to design appropriate fusion rules to generate a complementary output. To this end, numerous approaches have been developed in recent decades to design effective feature extraction strategies and appropriate fusion rules [20]–[24]. These approaches can be roughly classified into those based on a traditional framework and those based on deep learning.

Traditional approaches leverage multi-scale transformations (MST) [25], [26], sparse representation [27], [28], subspace decomposition, hybrid tools [29], mathematical optimization [23], [24], [30], [31] and others [32] to extract effective features. Among them, MST-based methods have attracted great attention due to their flexibility and advantages in great visual effects. These methods extract representative features at different scales by a specific transformation tool such as non-subsampled curvelet [33], wavelet [34], and edge-preserving filters [25]. Hence, the subsequent fusion process can fully embrace feature information on various scales and fuse them using simple max or averaging operators. However, this type of methods usually suffer halos and blurred edges caused by overlapping the unsymmetrical feature information at multiple scales.

Manuscript received August 31, 2020; revised November 28, 2020 and January 9, 2021; accepted January 21, 2021. Date of publication February 5, 2021; date of current version January 11, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2020YFB1313503; in part by the National Natural Science Foundation of China under Grant 61922019, Grant 61733002, Grant 61672125, and Grant 61772105; in part by the Liaoning Revitalization Talents Program under Grant XLYC1807088; and in part by the Fundamental Research Funds for the Central Universities. This article was recommended by Associate Editor L. K. Choi. (Corresponding author: Xin Fan.)

Jinyuan Liu and Ji Jiang are with the School of Software Technology, Dalian University of Technology, Dalian 116024, China, and also with the Key Laboratory for Ubiquitous Network and Service Software of Liaoning Province, Dalian University of Technology, Dalian 116024, China.

Xin Fan and Risheng Liu are with DUT-RU International School of Information Science, Dalian University of Technology, Dalian 116024, China, also with the Peng Cheng Laboratory, Shenzhen 518052, China, and also with the Key Laboratory for Ubiquitous Network and Service Software of Liaoning Province, Dalian University of Technology, Dalian 116024, China (e-mail: xin.fan@ieee.org).

Zhongxuan Luo is with School of Software Technology, Dalian University of Technology, Dalian 116024, China, also with the Peng Cheng Laboratory, Shenzhen 518052, China, and also with the Key Laboratory for Ubiquitous Network and Service Software of Liaoning Province, Dalian University of Technology, Dalian 116024, China.

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2021.3056725>.

Digital Object Identifier 10.1109/TCSVT.2021.3056725

1051-8215 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

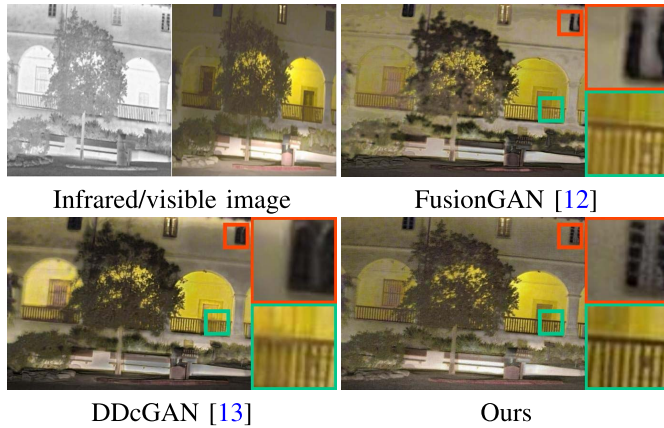


Fig. 1. Schematic illustration of the proposed method by making a comparison with the two state-of-the-art deep learning-based methods FusionGAN [12] and DDcGAN [13]. Obviously, our method simultaneously preserves the detail information and reduces unacceptable artifacts.

Recently, researchers employ deep learning (DL) for robust and efficient fusion of infrared and visible images [12], [35]–[37]. These DL-based methods achieve the state-of-the-art performance but still have limitations. First, the DL techniques are typically used to extract salient features from source images and then generate a weighting map for fusion [36], [38]. These deep salient features are given under one single scale neglecting the local/global information across scales, thus degrading the fusion quality to some extent [35], [39]. Second, these approaches take simple fusion rules, *e.g.*, addition and concatenation, so that undesirable artifacts or blurred edges may emerge in the final result. Last but not the least, training of deep networks demands a large number of *well-aligned* visible/infrared pairs that are difficult to collect in practice. FusionGAN [12] and DDcGAN [13] design generative adversarial networks (GAN) in order to enrich training examples for fusion. Nevertheless, the generated examples may lose textural details and destroy the intensity integrity of neighboring pixels as those GAN techniques randomly crop source images into non-overlapping image patches.

A. Contributions

This article presents a novel deep learning architecture for infrared and visible image fusion to overcome the limitations mentioned above. The proposed method learns both feature extraction and fusion rules, rendering robust and visually appealing results. Inspired by previous multi-scale feature ensemble methods [40]–[42], we design a densely contextualized dilated module in order to enlarge the receptive field, thus extracting deep salient features at multiple scales. The feature extractor strengthens the connections of feature maps at various scales without needing to change the size of source images, and meanwhile utilizes the multi-scale features more effectively by fully integrating the dense connections. It is worth noting that our training stage acquires *no* registered multi-modal image pairs, which expands the possibility to collect more training examples from either modal(s). In addition to the feature learning module, we develop an

edge-guided attention fusion rule in order to simultaneously preserve prominent image structures and suppress undesirable artifacts. Finally, we reconstruct the fused features by convolution operations with two skip connections from the feature extraction module to compensate for details loss. As a result, our method can generate more visually realistic fused results with rich textural details. Figure 1 illustrates a representative fusion example. The red and green boxes show that both FusionGAN [12] and DDcGAN [13] suffer from detail loss and blurred edges while our method does not but outputs better image contrast, leading to a distinctive target. We summarize our major contributions as follows:

- We propose a novel deep architecture for infrared and visible image fusion to learn salient features at multiple scales as well as fusion rules. Our method does not acquire registered image pairs in the training phase, which eliminates the dependence on a particular training dataset.
- In order to obtain more comprehensive features from the input images, we elaborately establish a feature extractor that takes a densely contextualized dilated network as its backbone to hierarchically integrate coarse-to-fine features from multiple scales. The afresh designed feature extractor effectively utilizes the intermediate features without the need of up or down-sampling source images.
- We formulate a cross-domain edge-guided attention mechanism to achieve data-consistency of the fused images with promising detail features, which preserves detailed information while attenuates noise or undesirable artifacts. This learning-based mechanism breaks through the limitations of manual-crafted fusion rules and remarkably improves the fusion performance.
- We construct a new aligned infrared and visible image fusion dataset, named RealStreet, with a broad range of challenging conditions including poor lighting and outdoor for performance evaluation.

Furthermore, extensive experiments on four datasets demonstrate the superiority of our method, outperforming state-of-the-art methods in terms of both subjective and objective evaluations by a large margin. Source code will be available at <https://github.com/JinyuanLiu-CV/MFEIF>.

II. RELATED WORKS

A. Traditional Approaches

A significant amount of traditional approaches have been developed for infrared and visible image fusion. They often rely on the spatial transformation techniques, such as wavelet transform [34], non-subsampled contourlet transform [33], contourlet transform [43], edge-preserving filter based transform [44] and Retinex theory-based transform [45]. For this purpose, Yan *et al.* [46] introduced a novel infrared and visible fusion method based on the spectral graph wavelet transform, and their method considered geometrical features with irregular regions of images. Moreover, Hu *et al.* [22] proposed an adaptive video-based method to deal with infrared and visible image fusion. The method separately generates two weighting maps through the cumulative distribution and entropy, and then these two maps guide the corresponding

layer to generate a fusion result. However, these methods typically lead to halos or undesirable artifacts in the fused result due to overlapping the information at different scales.

In addition to MST-based methods, SR-based methods learn an over-complete dictionary from a large number of natural images to construct sparse representation of original images. These methods commonly consist of three main steps: decompose input images into small patches as the preliminary for sparse coding; learn an over-complete dictionary from image patches; reconstruct the fused image from the sparse codes upon the learned dictionary. Liang *et al.* [47] introduced a method for multiple image fusion tasks by formulating a tensor and used singular value decomposition (SVD) to decompose the images. More recently, Kim *et al.* [48] introduced a novel dictionary learning method based on patch clustering and principal component analysis (PCA), achieving excellent fusion performance and removing redundancy of the learned dictionary. Although SR-based methods have achieved promising performance, the limitation is that only one well-trained dictionary cannot reflect the full information of input images.

Last, other fusion methods including Markov random fields [49], morphology and optimization model-based methods [11], [20] also shed light on image fusion. Ma *et al.* [11] introduced a total variation minimization model based on unregistered infrared and visible images, which can keep both thermal radiation of the infrared image and gradient of the visible image in the same scene. Nevertheless, the authors ignored the intensity information of the visible and the gradient of the infrared. To tackle this problem, Guo *et al.* [23] proposed a method based on augmented Lagrangian that improves the performance by considering both intensity distributions from multi-modal input images. Unfortunately, the optimization model can hardly characterize intrinsic image features underlying challenging multi-modal images.

B. Deep Learning Based Approaches

Deep networks have attracted great attention in image fusion during the last three years owing to their powerful representation ability on feature learning. Deep learning based methods can be broadly divided into three categories. The first group uses DL to extract salient features and generate weighting maps. Liu *et al.* [39] first used the convolutional neural network (CNN) for multi-focus image fusion task. The designed network is only applicable to multi-focus image fusion as it calculates binary maps. Li *et al.* [36] firstly decomposed source images into a base layer and a detail layer, and then generate weights upon the features extracted by deep networks. Finally, they apply the weights in the detail layer while simply average in the base layer for fusion. More recently, Favez Lahoud *et al.* [38] designed a zero learning framework for different image fusion tasks. They used pre-trained CNNs to generate weight maps for realizing different types of fusion tasks in a generic manner. The second group applies Generative Adversarial Networks (GANs). Ma *et al.* [12] used GAN to fuse the two type of images with a well-designed the content loss, which can simultaneously preserve the intensity from the infrared image and detail information in the visible image.

Xu *et al.* [13] designed dual-discriminators to enhance the target visibility by preserving the details from visible images and the discriminative target from infrared images. The third group is auto-encoder (AE) based fusion network. Li *et al.* trained an AE network to learn deep feature representation of multi-modal images. In the test stage, they merged feature maps with an appropriate fusion rule, and reconstructed an resultant image from the fused features through the trained decoder [35].

C. Attention Mechanism in DL

An attention mechanism that captures the region of interest originated from machine translation tasks and has been a hot topic in artificial intelligence [50]–[53]. Specifically, the mechanism uses an entire input sequence to calculate weights, and then introduces the weights into the input sequence to selectively pay more attention to important regions. It can also be explained by human biological systems that humans can easily observe important information and ignore other unimportant. The attention mechanism has played a key role in the image processing community (*e.g.*, saliency detection [54], image restoration [16], [55], [56], and semantic segmentation [57]). Wang *et al.* [58] introduced a network by stacking attention modules that can capture salient features, achieving outstanding recognition performance. Researchers also design edge-guided attention mechanisms in order to produce visually appealing images. Zhang *et al.* [59] proposed a novel method to improve the accuracy of medical image segmentation. They used an edge guidance module to learn edge-attention representation in early encoding layers, and then fused the transferred representation features by using a weighted aggregation module. Zhao *et al.* [60] introduced an edge guidance network (EGNet) to solve the coarse boundaries problems in object detection via the complementarity between salient edge and object information.

III. THE PROPOSED FUSION METHOD

This section gives the details of the proposed method including a coarse-to-fine feature extractor, edge-guided attention feature fusion, feature compensation reconstruction as well as the loss function. To begin with, the input infrared and visible images are separately fed into the feature extractor to acquire a series of intermediate features. Subsequently, the edge attention module generates attention maps that combine with the intermediate features for feature fusion. Finally, we reconstruct a final image from the fused features by convolutions with two skip connections from the feature extractor to further compensate the missing details.

A. Coarse-to-Fine Feature Extractor

One important issue for infrared and visible image fusion task is to extract rich features to represent input images. In general, feature extraction continues to push forward an immense influence on the fused result. Previous deep approaches designed a fully connected layer as the feature extractor without considering contextualized information,

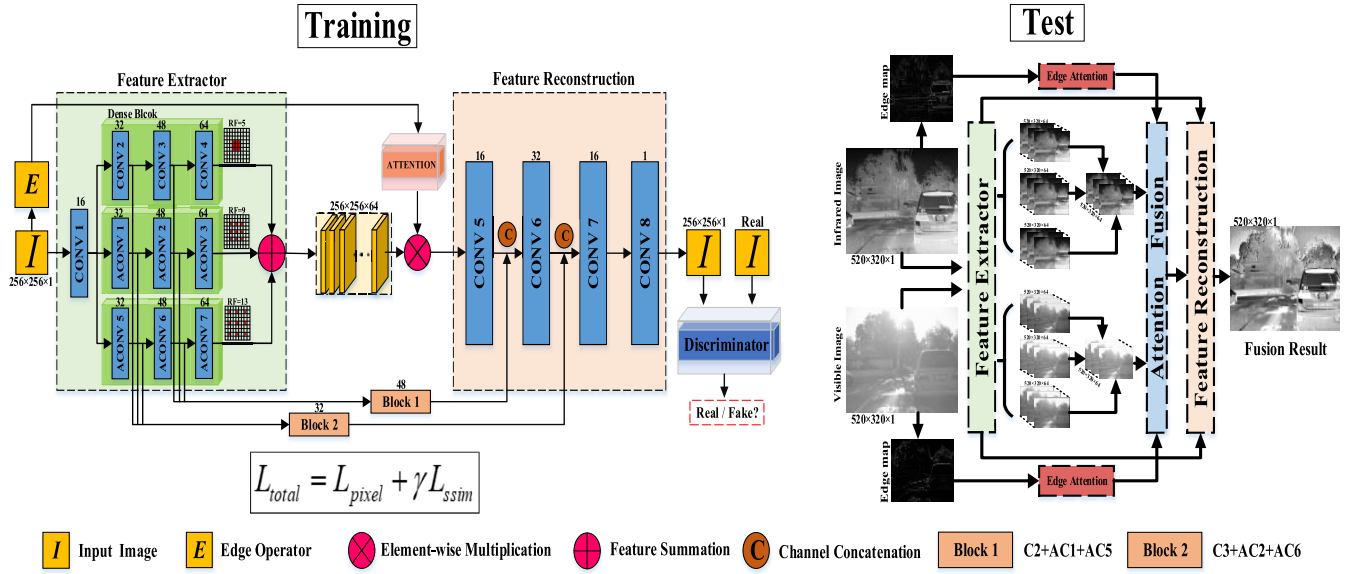


Fig. 2. Network architecture. In the training stage, we train the network in an end-to-end manner to learn a fixed coarse-to-fine feature extractor in addition to an edge guide attention mechanism. In the test stage, the trained model extracts the intermediate features of input images, and then generates edge attention maps to guide to fuse these extracted intermediate features, together with compensation features via two skip connections from the encoder part.

which may lead to undesired artifacts in the fused result. Therefore, we propose a contextualized dilated feature extraction module to obtain coarse-to-fine features in the following two ways: 1) we aggregate the representations of three convolution paths at multiple scales with different receptive fields by a multi-scale context aggregation structure, and 2) we integrate a dense block in each convolution path to provide richer features for subsequent use.

As shown in Fig. 2, the network transforms the infrared and visible images into the feature space via the first convolution. Then, the intermediate results from the three convolutional paths with different dilated factors are aggregated to acquire overall information at multiple scales. The dilated convolution [61] uses the step size of the magnifying factor to weigh pixels, and thus increases its receptive field without changing resolution. Each dilated path consists of three convolutions by using the same kernel size 3×3 . The three paths use their typical receptive fields, 5×5 , 9×9 , and 13×13 , respectively, to provide more precisely complementary information.

Let $\mathbf{f}_{in}$ denote the input feature map for the extraction module, and we calculate the output feature map $\mathbf{f}_{out}^e$ of the extraction model as follows:

$$\mathbf{f}_{out}^e = \max(0, \sum_{t_p=1}^3 (\mathbf{W}_{t_p} * \mathbf{f}_{in, t_p} + \mathbf{b}_{t_p})), \quad (1)$$

where $*$ denotes the convolution operator and t_p expresses the sequence number of dilated convolution paths. $\mathbf{W}$ and $\mathbf{b}$ represent the filter parameter and basis of the convolution layer, respectively.

In addition to aggregating the salient features at multiple scales, we further add a dense connection in each dilated path to preserve deep features as much as possible. The output of each layer is cascaded as the input of the next layer. We denote

the output features of the infrared and visible images given by the feature extraction module as $\mathbf{f}_{ir}^e$ and $\mathbf{f}_{vis}^e$, respectively.

The densely dilated feature extraction module adequately integrates the information from different receptive fields and ensures the structural consistency of deep features.

B. Edge-Guided Attention Feature Fusion

A variety of attention mechanisms have been successfully applied in many computer vision tasks since they can capture the region of interest in visual scenes. The major objective of fusion is to find the appropriate characteristics of each modality. For this purpose and motivated by previous works, we utilize rough intermediate features to obtain the attention maps of enhanced edge images. Through this designed edge-based attention mechanism, our fusion results can simultaneously preserve more textural details and attenuate undesirable artifacts.

Specifically, the edge map is obtained in two steps. We denote an input gray-scale image with the size $m \times n$ as $\mathbf{u}$, and define its gradient map $\nabla \mathbf{u}$ as:

$$\nabla \mathbf{u} = \sum_{i=1}^{mn} \sqrt{(\nabla_i^h \mathbf{u})^2 + (\nabla_i^v \mathbf{u})^2}, \quad (2)$$

where $\nabla_i^h \mathbf{u} = \mathbf{u}_i - \mathbf{u}_{a(i)}$ and $\nabla_i^v \mathbf{u} = \mathbf{u}_i - \mathbf{u}_{b(i)}$ are linear operators to calculate the horizontal and vertical first-order differences, respectively. The labels $\mathbf{u}_{a(i)}$ and $\mathbf{u}_{b(i)}$ represent the nearest neighbor pixels that locate at the right and below the source pixel i , respectively. We also design an edge-enhanced operator $\mathcal{S}$ to make gradient information more remarkable:

$$\mathcal{S}(\nabla \mathbf{u}) = \max_{j \in J} (\max_{i \in I} \nabla \mathbf{u}(i+1, j+1), \nabla \mathbf{u}(i, j)), \quad (3)$$

where $I = \{1, \dots, m-1\}$ and $J = \{1, \dots, n-1\}$. The indices i and j represent the horizontal and vertical directions of the gradient image, respectively.

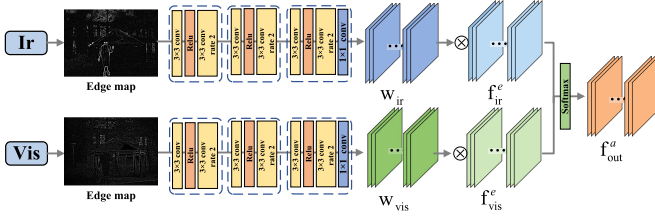


Fig. 3. The procedure of edge-guide attention mechanism. *rate 2* denotes dilated convolution operator with a dilation rate of two.

Subsequently, we feed the enhanced edge maps of infrared and visible images into the attention mechanism to generate feature weight maps $\mathbf{W}_{ir}$ and $\mathbf{W}_{vis}$, and calculate the fused feature $\mathbf{f}_{out}^a$ weighted by edge-guided attention:

$$\mathbf{f}_{out}^a = \text{softmax} \left(\sum_{i=1}^k (\mathbf{f}_{ir}^e \mathbf{W}_{ir} + \mathbf{f}_{vis}^e \mathbf{W}_{vis}) \right). \quad (4)$$

The summation of two-modal features $\mathbf{f}_{ir}^e$ and $\mathbf{f}_{vis}^e$ from the extraction module weighted by the attention maps $\mathbf{W}_{ir}$ and $\mathbf{W}_{vis}$ generates the fused feature as shown in Fig. 3. The final fused image is reconstructed from the fused feature $\mathbf{f}_{out}^a$ by the trained decoders as given below.

C. Feature Compensation Reconstruction

Image reconstruction aims at transforming feature maps from the feature space to image space by convolution layers. Simply utilizing convolution operations may lead to vital information loss in the process of recovering images. We introduce two skip connections to alleviate the information loss of feature maps after multiple convolutions.

Specifically, we sum three different dilated features from the feature extractor module to compensate the infrared and visible features, and then use the choose-max strategy to select either the infrared or visible features in an element-wise manner. Finally, we concatenate these compensated features to the attention-based fused features along channels in the feature reconstruction module.

As a consequence, the fused image is recovered from the feature reconstruction module. The flowchart of our method is illustrated in Fig. 2.

D. Loss Function and Training Details

For multi-modal image fusion, there is no ground truth for networks to execute supervised/unsupervised learning. Besides, it is difficult to obtain sufficient registered image pairs in a real-world scenario. To this end, we train our networks by feeding an infrared or a visible, by which we can reconstruct the input image via the encoder-decoder part, in addition to the edge attention mechanism. The detailed framework of our method in the training phase is shown on the left side of Fig. 2.

To reconstruct the input image more precisely, we minimize the total loss function $\mathcal{L}_{total}$ (the combination of $\mathcal{L}_{SSIM}$ and $\mathcal{L}_{MSE}$ with a hyperparameter γ) to train our network. SSIM [62] is an effective measure of structural similarity between two different images, which combines three

components, *i.e.*, luminance, structure, and contrast. Meanwhile, MSE targets to measure the pixel intensity between the input and output images. These two loss functions together constrain the reconstructed structure and pixel errors. The total loss function is expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{MSE} + \gamma \mathcal{L}_{SSIM}. \quad (5)$$

$\mathcal{L}_{MSE}$ calculates the Euclidean distance between the input and output images:

$$\mathcal{L}_{MSE} = \sqrt{\frac{1}{MN} \sum_{x \in M, y \in N} (out(x, y) - in(x, y))^2}, \quad (6)$$

where *out* and *in* indicate the reconstructed data and input training data, respectively. *M* and *N* give the size of an image and (x, y) is the pixel location. $\mathcal{L}_{SSIM}$ is calculated as:

$$\mathcal{L}_{SSIM} = 1 - SSIM(out, in). \quad (7)$$

The symbol $SSIM(\cdot)$ represents the structural similarity operation.

Furthermore, inspired by generative adversarial networks (GANs) [63], [64], we suppose the aforementioned network as a generator and add a discriminator at the end of the network to guide the generator produce more natural images. The added $\mathcal{L}_{Adv}$ can be written as:

$$\min_G \max_D \mathcal{L}_{Adv}(G, D) = \mathbb{E}[\log D(out)] + \mathbb{E}[\log(1 - D(G(in)))]. \quad (8)$$

The generator and discriminator are alternative iterations, rendering a more robust and reliable network.

At the end of training, we feed two registered images in our network. The trained coarse-to-fine model is utilized to extract deep salient features from input two-modal images. Subsequently, the jointly edge-guided weight maps are exploited to multiply the corresponding features, generating fused features. Finally, the fused features are fed back into the decoder module with two added skip connections to reconstruct the final fused image.

In our training phase, we separately train the proposed network on the MSCOCO [65] and FLIR datasets,¹ respectively. The MS-COCO dataset only contains visible images with low spatial resolution, while the FLIR dataset contains both infrared and visible images of each scene. We select 20,000 images as training images and 1500 images as validation data on the MS-COCO dataset. Comparably, we choose 10,000 infrared and 10,000 visible images as training, together with 1,000 infrared images and 1,000 visible images as validation data on the FLIR dataset. We conduct ablation experiments towards the proposed loss function on these two training datasets. The validation of total loss on two datasets and visual comparison results towards with/without $\mathcal{L}_{Adv}$ loss are demonstrated in Figs. 4 and 5, respectively. We can see that the total loss value tends to be stable in each case. The total loss on the FLIR dataset pushes faster than that on the

¹<https://www.flir.ca/oem/adas/adas-dataset-form/>

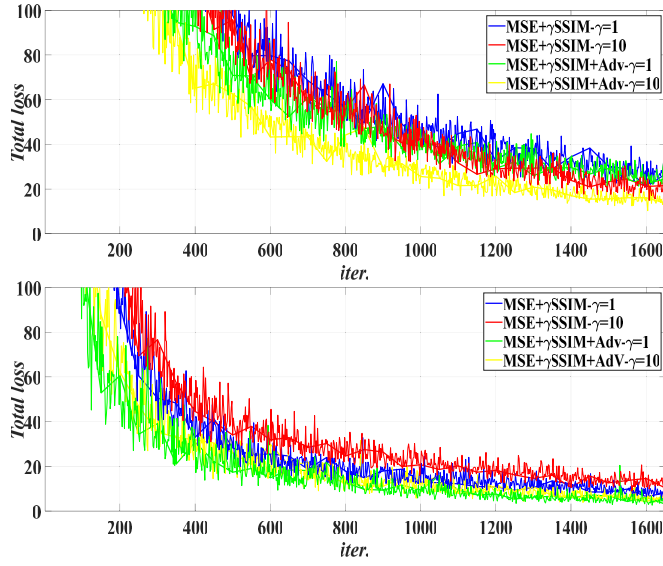


Fig. 4. The graph plot of total loss w/o $\mathcal{L}_{Adv}$ by using different training datasets. We train the proposed network on MSCOCO dataset in the top row and FLIR dataset in the bottom row, respectively. The vertical axis denoted loss value, while the horizontal axis indicates iterations.

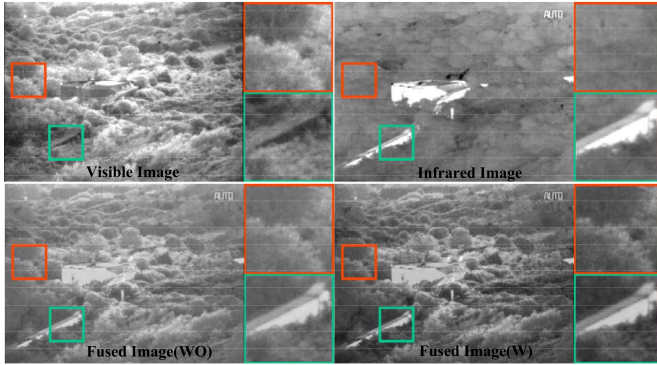


Fig. 5. From left to right: visible image, infrared image, our method without discriminator, our fused result with discriminator. The discriminator guided the fused result closer to a natural image.

MS-COCO dataset, due to the high quality multi-modality images. Besides, the designed discriminator further converges the total loss to a smaller value. Input images are randomly cropped to size 256×256 and transformed to grayscale images. We train the network by the Adam optimizer with a learning rate of 1×10^{-4} . The batch size and epoch are sets to 16 and 4, respectively. Our network is implemented on TensorFlow and trained on a PC with an NVIDIA GTX 1080Ti (GPU), 32GB RAM (Memory), and Intel Core i7-8750 (CPU).

IV. EXPERIMENTS

A. Image Fusion Metrics

In most cases, the subjective visual perception system is susceptible to human factors, such as personal emotion, eyesight. Also, the fused results among different methods always bring some resemblance. Here, for quantitative comparison between our fusion framework and other existing approaches, six evaluation metrics including

mutual information (MI) [66], correlation coefficient (CC), visual information fidelity (VIF) [67], edge-based similarity measure ($Q^{AB/F}$) [68], structural similarity index measure (SSIM) [62], and the sum of the correlations of differences (SCD) [69] are utilized.

The mutual information (MI) evaluation metric derived from information theory, which measures the amount of information that is transferred from the input images to the final fused result. MI is calculated as follows:

$$MI = MI_{A,F} + MI_{B,F}, \quad (9)$$

where $MI_{A,F}$ and $MI_{B,F}$ denote the total amount of information that is transferred from multi-modal input images to the final fused result. The two random variables in MI is calculated by the Kullback-Leibler measure:

$$MI_{X,F} = \sum_{x,f} p_{X,F}(x,f) \log \frac{p_{X,F}(x,f)}{p_X(x)p_F(f)}, \quad (10)$$

where $p_{X,F}(x,f)$ denotes the joint histogram of input image X and the fused result F . The $p_X(x)$ and the $p_F(f)$ calculate the marginal histograms of input image X and the fused result F . Usually, a large MI indicates that the abundant information transferred from inputs to the fusion result and also represents good fusion performance.

The correlation coefficient (CC) measures the degree of linear correlation between the input images and the fused result. Mathematically, CC is defined as follows:

$$CC = \omega_a r_{af} + \omega_b r_{bf}, \quad (11)$$

where $\gamma_{xf} = \frac{\sum_{i=1}^M \sum_{j=1}^N (x_{i,j} - \mu_x)(f_{i,j} - \mu_f)}{\sqrt{\sum_{i=1}^M \sum_{j=1}^N (x_{i,j} - \mu_x)^2 \sum_{i=1}^M \sum_{j=1}^N (f_{i,j} - \mu_f)^2}}$, μ_x and μ_f denote the mean values of the input image x and the fusion result y , respectively. A higher CC value indicates that the fused result is more similar to the source images and also presents a good performance. Structural similarity index measure (SSIM) is a universal quality index that is widely used to model image loss and distortion. The index mainly consists of loss of correlation, light, and contrast distortion. The SSIM between the input X and the output F is defined as follows:

$$SSIM_{X,F} = \sum_{x,f} \frac{2\mu_x\mu_f + C_1}{\mu_x^2 + \mu_f^2 + C_1} \cdot \frac{2\sigma_x\sigma_f + C_2}{\sigma_x^2 + \sigma_f^2 + C_2} \cdot \frac{\sigma_{xf} + C_3}{\sigma_x\sigma_f + C_3}, \quad (12)$$

where μ and σ denote the mean value and standard covariance, respectively. C_1 , C_2 and C_3 are the parameters to ensure the algorithm robust and stable. Therefore, the structural similarities between the source images A, B and the fused result F can be written as:

$$SSIM = SSIM_{A,F} + SSIM_{B,F}. \quad (13)$$

The VIF mainly computes the information fidelity of a fused image, which is in line with human visual perception. VIF targets to establish a model to calculate the distortion between the inputs and fused result. The larger VIF is in better correspondence with the response of the human visual system.

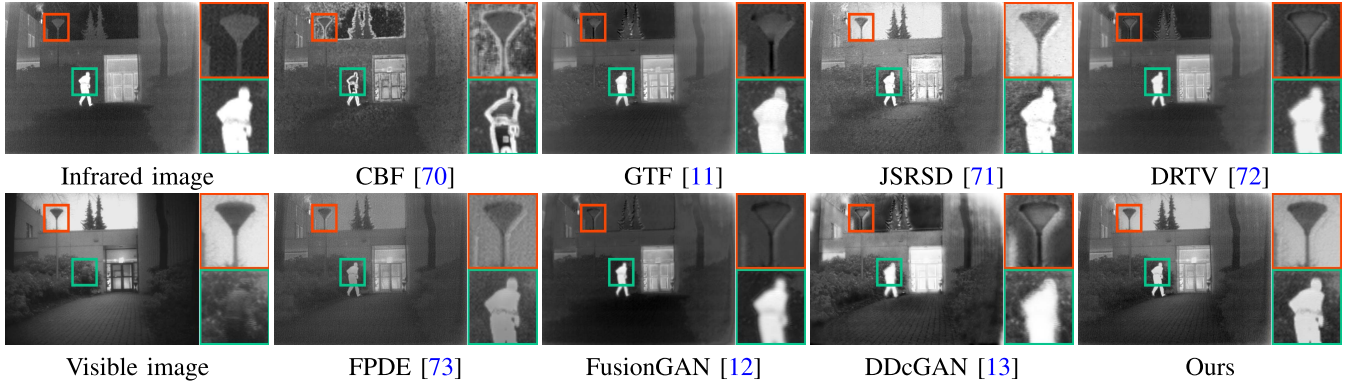


Fig. 6. Visual results comparison between different methods of Kaptein-1123 from the TNO dataset. Our method is better in preserving detailed textural details, especially in the zoomed-in patches. Best viewed on screen.

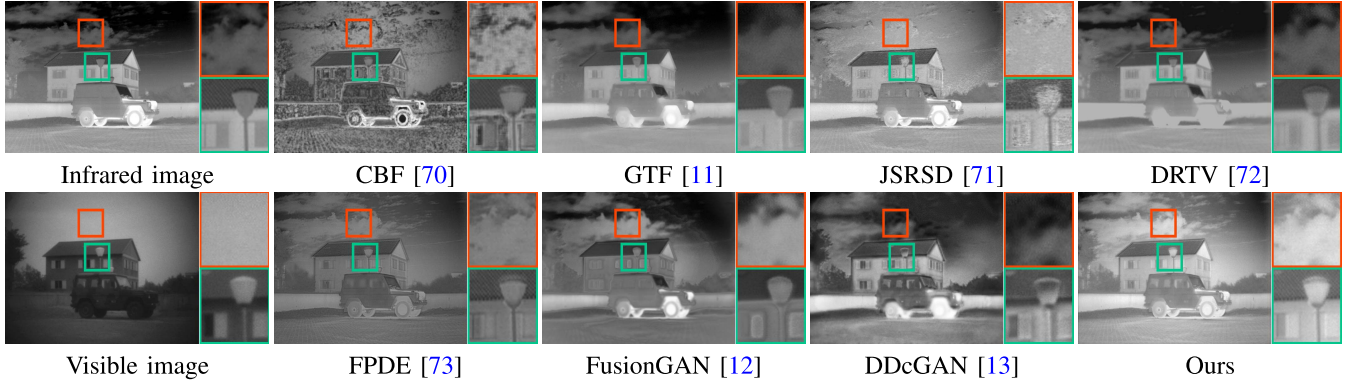


Fig. 7. Visual results comparison between different methods of Marne_04 from the TNO dataset. It can be seen clearly that only our method preserves the cloud without noises as shown in the red frame. Best viewed on screen.

$Q^{AB/F}$ applies a statistical method to measure the total edges information transferred from the source images to the fusion result, and it defined as follows:

$$Q^{AB/F} = \frac{\sum_{i=1}^N \sum_{j=1}^M (Q^{AF}(i, j)w_A(i, j) + (Q^{BF}(i, j)w_B(i, j)))}{\sum_i \sum_j (w_A(i, j) + w_B(i, j))}, \quad (14)$$

where N and M are height and weight of the image. $Q^{AF}(i, j) = Q_g^{AF}(i, j)Q_o^{AF}(i, j)$, $Q_g^{AF}(i, j)$ and $Q_o^{AF}(i, j)$ are the edge strength and orientation preservation value at the location i, j , $w_A(i, j)$ and $w_B(i, j)$ represent the saliency weight of $Q^{AF}(i, j)$ and $Q^{BF}(i, j)$, respectively. A large $Q^{AB/F}$ indicates that considerable edge information is transfer to the fused result.

SCD aims to measure how much information is transmitted from input images to the fused image. It can be formulated as:

$$SCD = r(D_1, S_1) + r(D_2, S_2), \quad (15)$$

where $r(\cdot)$ represents the correlation function which is calculate as:

$$r(D_k, S_K) = \frac{\sum_i \sum_j (D_k(i, j) - \bar{D}_k)(S_k(i, j) - \bar{S}_k)}{\sqrt{(\sum_i \sum_j (D_k(i, j) - \bar{D}_k)^2)(\sum_i \sum_j (S_k(i, j) - \bar{S}_k)^2)}} \quad (16)$$

where $k = 1, 2$ and $\bar{S}_k$ and $\bar{D}_k$ represent the average pixel values in terms of S_k and D_k . A positive value of SCD means a good fusion quality.

B. Results Analysis on TNO Dataset

We evaluate the performance of our method by making a comparison with nine state-of-the-art methods on TNO dataset² including transform based-method, *i.e.*, FPDE [73] and CBF [70], total variation-based method, *i.e.*, GTF [11] and DRTV [72], sparse coding-based method JSRSD [71] and two deep learning-based method, *i.e.*, FusionGAN [12] and DDcGAN [13].

Qualitative Comparisons For qualitative comparison, we firstly perform qualitative experiments. The intuitive results are shown in Fig.6 and Fig.7. Through comparing with other existing methods, it can be noted that our method has three significant advantages. Firstly, our method can preserve abundant details. As shown in Fig. 7, the contour of the car can be seen much more clearly with the proposed method. Three different groups of dilated convolution are adopted to extract features, making sure all the multi-scale information is acquired. Secondly, our results have less undesirable artifacts which are presented in Fig. 6. Although FusionGAN and DDcGAN suppress noise to some extent, only our method presents proper contrast. It is benefited from the attention mechanism, which can avoid generating undesirable artifacts and preserve more small textural details. Lastly, our results

²https://figshare.com/articles/TNO_Image_Fusion_Dataset/1008029

TABLE I

THE QUANTITATIVE RESULTS OF THREE SEQUENCE DATASETS ON THE TNO DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset:TNO									
	Metrics	CBF	GTF	JSRSD	DRTV	FPDE	FusionGAN	DDcGAN	Ours
Nato	MI	1.430	1.990	1.478	2.461	1.309	2.422	2.340	2.510
	CC	0.578	0.461	0.589	0.453	0.570	0.500	0.670	0.670
	VIF	0.302	0.357	0.171	0.276	0.313	0.215	0.430	0.401
	$Q^{AB/F}$	0.342	0.371	0.197	0.252	0.252	0.197	0.220	0.390
	SSIM	0.413	0.439	0.253	0.407	0.365	0.349	0.350	0.563
	SCD	1.340	0.960	1.560	0.883	1.132	1.122	1.010	1.770
Tree	MI	1.372	1.675	1.606	0.602	1.814	1.623	2.190	2.591
	CC	0.635	0.608	0.594	0.276	0.744	0.615	0.660	0.702
	VIF	0.259	0.414	0.196	0.063	0.432	0.254	0.520	0.530
	$Q^{AB/F}$	0.208	0.349	0.157	0.051	0.421	0.178	0.271	0.310
	SSIM	0.296	0.442	0.227	0.104	0.493	0.228	0.370	0.503
	SCD	1.201	0.701	1.210	0.204	1.122	0.960	1.045	1.670
Duine	MI	1.264	1.235	0.805	1.699	1.332	1.767	1.671	1.931
	CC	0.678	0.522	0.551	0.523	0.701	0.528	0.280	0.720
	VIF	0.365	0.463	0.157	0.350	0.431	0.378	0.410	0.510
	$Q^{AB/F}$	0.342	0.371	0.197	0.252	0.412	0.197	0.220	0.390
	SSIM	0.521	0.524	0.176	0.501	0.540	0.352	0.360	0.531
	SCD	1.250	0.730	1.210	0.672	1.321	1.640	0.991	1.931

contain more salient detail information, the visual result not only shows the external information of the car but also the internal of it in Fig.7. In addition, our method can restore the shape of the cloud completely.

Quantitative Comparisons In addition to visual results, we employ six evaluation matrix, including mutual information (MI) [74], correlation coefficient (CC) [75], visual information fidelity (VIF) [67], $Q^{AB/F}$ [68], the sum of the correlations of differences (SCD) [69] and structure similarity (SSIM) [62], to further verify the effectiveness of the proposed method. All these experiments based on the 25 image pairs and three video datasets on the TNO dataset. Fig.8 shows the mean and variance value of six evaluation metrics on the TNO dataset. The node denotes the mean value on 20 test image pairs, while the length of the line represents variance value (fluctuation range). As can be seen from the statistical results, our method generates the largest mean values in terms of five evaluation metrics: MI, CC, VIF, SSIM, SCD. For the metric $Q^{AB/F}$, our method achieves comparable results with the average value being the second largest. Meanwhile, achieving a lower variance indicates that our method is more stable in dealing with various visual scenes. Specifically, MI and $Q^{AB/F}$ comprehensively prove that our method transfers more considerable information from the source images. Values of VIF and SCD denote that our fused results are more friendly to human visual system with abundant texture details. Besides, the highest value of CC and SSIM reveals that our fused results best continue the structure and vital information from the input images. Table I reveals that our method can achieve stable and excellent quantitative results in different visual scenes. These results demonstrate our method can transfer more considerable information from source images, especially the richest texture details and the distinct contrast.

C. Results Analysis on Real-Street Dataset

We collected a dataset outdoors during the night, named Real-Street. It consists of 40 visible-infrared pairs, including

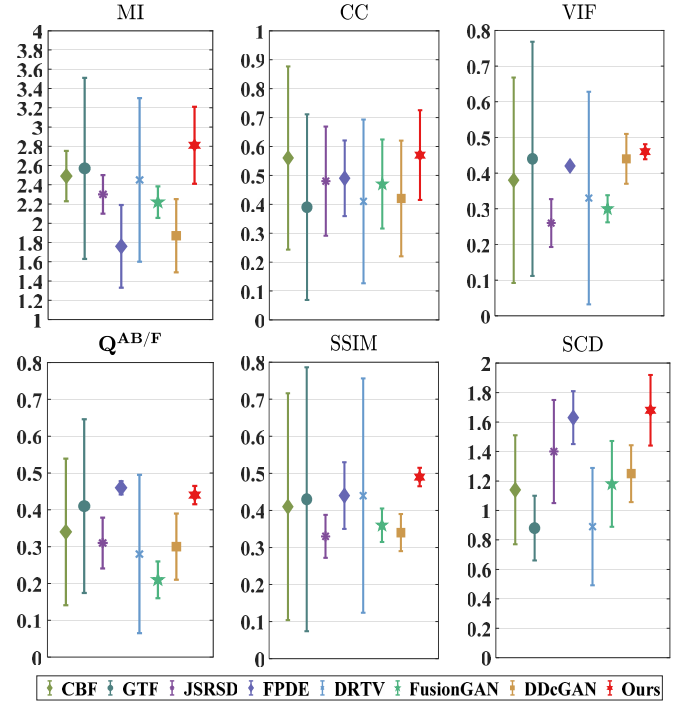


Fig. 8. Quantitative comparison of six metrics for 20 fused images on the TNO dataset. The nodes represent average value and the length of lines means the fluctuation range.

a pedestrian sequence with 33 image pairs and 7 independent street scenes. The pairs of infrared and visible images showing on the existing available dates were presented in low resolution, especially, a great deal of the images were not in line with the human visual system. In order to overcome the fusion problem well, we employed two cameras that have high resolution to ensure the high-quality of inputs. It sheds light on fusing the high-quality infrared and visible image and makes up for the shortcoming of the existing datasets. Additionally, our collected images are registered through the parameters of

TABLE II
THE QUANTITATIVE RESULTS ON REALSTREET DATASET FROM TNO. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Metrics	Dataset: RealStreet							
	CBF	GTF	JSRSD	DRTV	FPDE	FusionGAN	DDcGAN	Ours
MI	3.140	3.241	1.823	3.161	2.812	2.532	2.508	3.615
CC	0.687	0.731	0.619	0.724	0.733	0.703	0.722	0.745
VIF	0.415	0.456	0.188	0.406	0.392	0.217	0.405	0.421
$Q^{AB/F}$	0.577	0.582	0.324	0.528	0.522	0.303	0.485	0.422
SSIM	0.423	0.452	0.233	0.447	0.475	0.263	0.392	0.553
SCD	0.951	1.511	0.817	1.532	1.121	0.687	1.411	1.377

TABLE III
TAVERAGE RUNTIME COMPARISON OF DIFFERENT METHODS ON THE TWO DATASETS (UNIT: SECOND). AS FOR THE RUNTIME OF DEEP LEARNING-BASEDMETHOD, THE VALUE IS TEST ON GPU

	Dataset	CBF	GTF	JSRSD	DRTV	FPDE	FusionGAN	DDcGAN	Ours
Time	TNO	90.42	4.05	640.21	2.87	5.07	0.75	1.54	1.42
	RealStreet	58.55	2.71	396.27	1.86	3.25	0.53	1.03	0.95
Param/FLOPS	TNO	-/-	-/-	-/-	-/-	-/-	925k/56.3G	1097k/311G	158k/25.3G

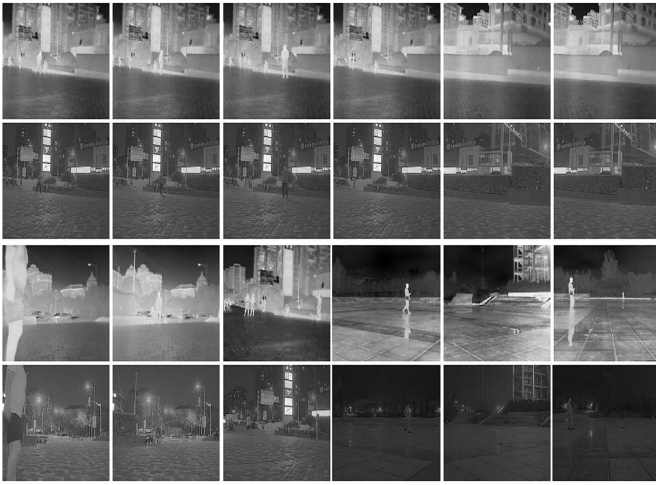


Fig. 9. Examples of the infrared and visible image pairs on the proposed RealStreet Dataset. The first and the third row contains visible images, while the second and the forth row presents corresponding infrared images.

the two cameras. The samples of the proposed dataset are shown in Fig. 9.

Visual comparisons of different fusion methods are shown in Fig. 10, it can be seen that undesired artifacts arise in CBF and JSRSD. While for GTF, DRTV, these methods depend on the using gradient for detail information modeling, the details of an infrared image have not been recovered in the fused results, leading to the loss of abundant information. Additionally, FusionGAN and DDcGAN produce under-exposed results, the person and the buildings are not noticeable on the fusion result. In contrast, the proposed method considers both infrared and visible information. The structure of infrared images and the details of visible images are taken into consideration simultaneously to generate visual-friendly fused results. The quantitative comparisons between our method and other state-of-the-art approaches on Real-Street dataset are listed in Table II. Our method runs steadily and achieves the largest average values on three metrics including MI, CC, and SSIM. These results reveal that our results own strongly correlated information and structure with the two source images. However, our method ranks second behind GTF on VIF. This is because that GTF uses infrared images

TABLE IV
ABLATION ANALYSIS OF DIFFERENT MODULE INTEGRATION. $\mathcal{D}$ REPRESENTS DENSELY DILATED FEATURE EXTRACTION MODULE, $\mathcal{A}$ REPRESENTS THE EDGE-GUIDED ATTENTION MECHANISM AND $\mathcal{S}$ EXPRESSED AS SKIP CONNECTION

Method	MI	CC	VIF	$Q^{AB/F}$	SCD	SSIM
Ours w/o $\mathcal{D}$	2.65	0.59	0.34	0.34	1.61	0.35
Ours w/o $\mathcal{A}$	2.42	0.67	0.44	0.40	1.50	0.41
Ours w/o $\mathcal{S}$	2.28	0.65	0.42	0.41	1.44	0.42
Ours	2.83	0.65	0.49	0.53	1.65	0.45

as the data-fidelity term to establish a TV fusion model. Since infrared images have great advantages in representing the structure information and contrast of the image, GTF can achieve considerable VIF and $Q^{AB/F}$ value when the quality of input infrared images are high. In contrast, our method more balances structural information. As a result, our method has a slightly lower $Q^{AB/F}$ value than other methods (e.g., CBF, GTF, and FPDE).

D. Time Complexity Analyse

The TNO dataset consists of images of different size, while images on the RealStreet dataset share the same size of 512×384 . Therefore, for a more objective evaluation, the time complexity analysis experiment is tested on the RealStreet Dataset. Table III indicates that JSRSD has high computational complexity. For traditional-based methods, including CBF, GTF, DRTV, and FPDE, which contain complex scale transform or a large number of iterations, which leads to time-consuming. In contrast, due to the strong computing ability of CNNs, deep learning based methods, e.g., FusionGAN, DDcGAN, and our method achieve high speed when fusing images. It worth pointing out that our method has lower computing complexity in terms of FLOPs and fewer training parameters, achieving comparable efficiency compared to the other methods.

E. Ablation Study

To verify the validity of each module in the proposed method, the quantitatively ablation study is proved in Tab.IV. In Table IV, $\mathcal{D}$ represents densely dilated feature extraction module, $\mathcal{A}$ represents the edge-guided attention mechanism

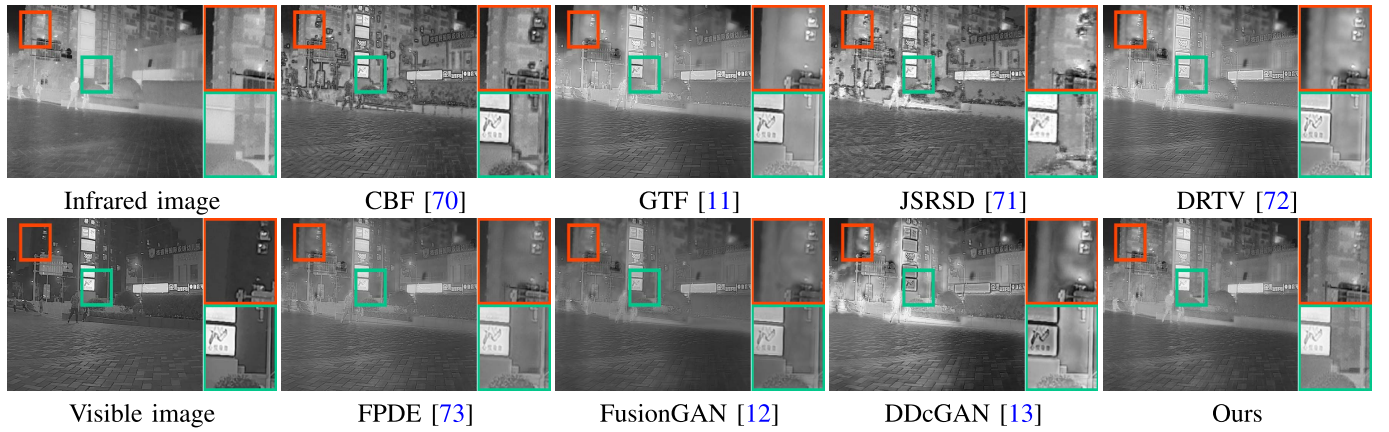


Fig. 10. Visual results comparison between different methods on the Real-Street dataset. Clearly, only our method can preserve both the thermal radiation and the texture information, e.g., building in the red frame. Best viewed on screen.

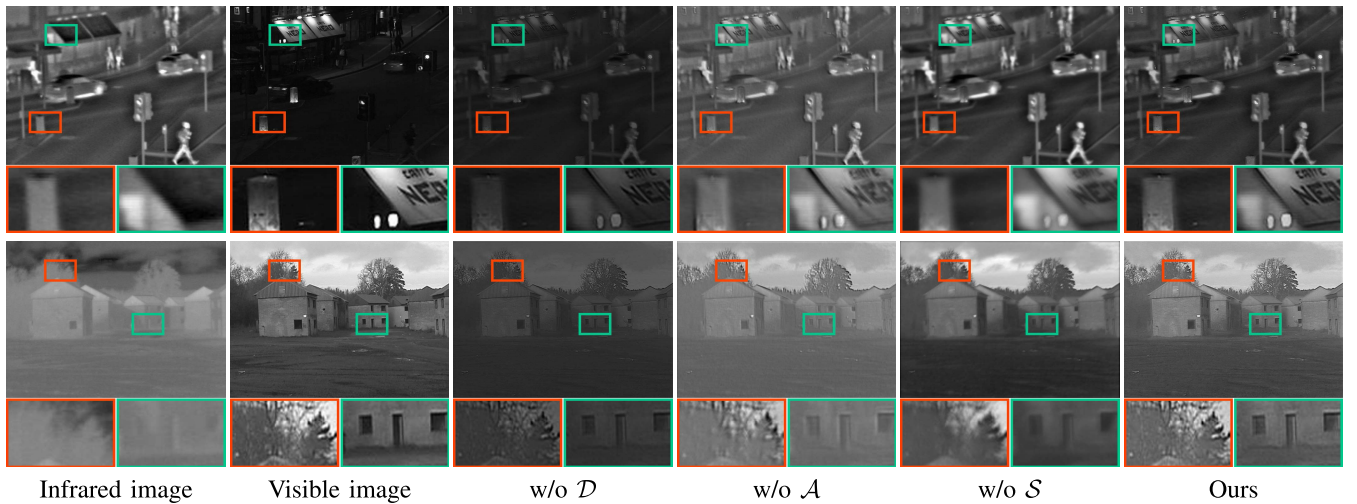


Fig. 11. Ablation study about the effect of each module of two typical images from the TNO dataset. Each module plays a key role in generating a visual-friendly and noise-free fusion result.

and $\mathcal{S}$ expressed as skip connection. In order to verify the effectiveness of these modules, we separately conducted ablation experiments on the three modules, and compared the performance with/without it. As shown in Table IV, MI drops significantly without module $\mathcal{S}$, since module $\mathcal{S}$ makes compensation for information lost in the convolution layer, and the features of it can be reused in the reconstruction. That is to say, module $\mathcal{S}$ is the most important module in terms of MI. Besides, module $\mathcal{D}$ is a coarse-to-fine feature extractor, which plays a key role in extracting salient features on multiple scales. Both VIF and SSIM are sensitive to module $\mathcal{D}$, which implies that the module $\mathcal{D}$ is more powerful for preserving the consistent structure of the image. As the global information of the image is fully extracted by module $\mathcal{D}$, the fusion result is effectively guaranteed. Apart from that, although metric CC achieves the top score without module $\mathcal{A}$, the other five metrics all significantly decline to a certain degree. That is because module $\mathcal{A}$ concentrates on calculating the salient detail information from the input images, which may lose some relevant information in the processing. In general, each

module in our network positively impacts on the fusion result, demonstrating that the proposed method is reasonable and effective.

We also demonstrate the fusion result of each module in Fig.11. Each module plays a role in generating high-quality results. It can be easily noticed that without the densely dilated feature extraction module, the fusion results tend to be low contrast, due to the fact that intermediate features in the network are not fully utilized. Additionally, the attention fusion mechanism guides the intermediate features to learn more detail information from the input images, leading to enhance small textural details and suppress the noises. Last but not the least, the skip connection further compensates for the vital information lost from the encoder part, ensuring all the vital information transfers to the decoder part.

Finally, we explore the role of each contextualized dilated layer in the proposed method. The visual results are shown in Fig. 12. It can be seen that the increasingly larger receptive field takes every effort to preserve the vital intermediate information from coarse to fine, which is beneficial to provide

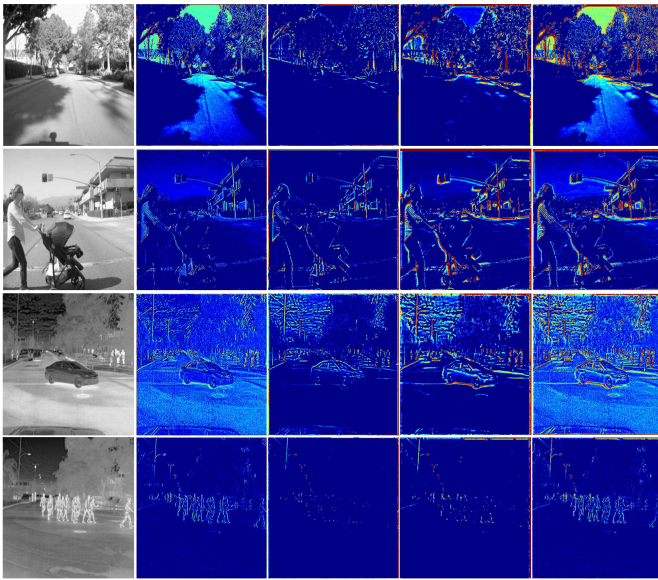


Fig. 12. From left to right: input infrared or visible images, dilated convolution with receptive fields 5×5 , dilated convolution with receptive fields 9×9 , dilated convolution with receptive fields 13×13 , as well as the final output of dilated feature extraction module.

overall information to improve the fusion result. The experimental results demonstrate the effectiveness of each densely contextualized dilated convolutional layer.

F. Extension to Multiple Datasets

In this section, a fusion scheme for color image fusion is proposed. As we all know, the HSV color space is less sensitive to brightness. The V component determines the brightness of the color (referred to as luminance), while the H and S components determine the intrinsic color. We transfer the input visible images from RGB color space to HSV color space, and then fuse V component and infrared/near-infrared with fixed H and S components to obtain a new fused V component, replacing the foregoing V component. In the end, we transfer the fused image from HSV color space to RGB color space.

To further verify the generalization of the proposed method, we further conducted the experiment on two other datasets. In total, there are 40 pairs of infrared and visible images from FLIR dataset, and 30 pairs of near-infrared and visible images from NIR dataset [76] for testing the proposed method, which contains daylight and nightlight illuminations. For FLIR dataset, the typical image pairs are shown in Fig.13. GTF, FusionGAN and DDcGAN lose vital detail information, especially in the green frame. Additionally, JSRSD suffered from noise interference because the learned dictionary cannot express all the unseen pixel. Only CBF and the proposed method retain the detail information from the input images. However, the sky in CBF still exists undesirable artifacts.

Near-infrared (NIR) is an electromagnetic wave between visible light and medium infrared light, which improves the overall visibility of the scene on various occasions. Fig.14 shows the visual comparison of the proposed method and the

TABLE V
THE QUANTITATIVE RESULTS ON FLIR AND NIR DATASETS.
AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Metric	CBF	GTF	JSRSD	FusionGAN	DDcGAN	Ours
FLIR	MI	3.15	3.27	2.25	2.74	2.75	2.96
	CC	0.70	0.69	0.73	0.66	0.72	0.80
	VIF	0.46	0.41	0.23	0.33	0.40	0.47
	$Q^{AB/F}$	0.52	0.31	0.24	0.25	0.25	0.44
	SSIM	0.41	0.43	0.25	0.53	0.35	0.55
	SCD	0.978	0.92	1.32	1.08	1.41	1.49
NIR	MI	3.61	3.87	2.11	2.32	3.01	4.28
	CC	0.80	0.78	0.77	0.79	0.76	0.87
	VIF	0.42	0.52	0.18	0.41	0.44	0.45
	$Q^{AB/F}$	0.47	0.49	0.21	0.35	0.33	0.44
	SSIM	0.44	0.53	0.22	0.40	0.37	0.55
	SCD	0.40	0.13	0.45	0.50	0.57	1.12

five state-of-the-art methods. The visible image shows low visibility on the background plain and mountain due to the haze interference. In contrast, the NIR image presents significant details in the region. Both GTF and JSRSD are over-exposed. Apart from that, the halos and undesirable artifacts are emerging on the results of FusionGAN and DDcGAN, due to excessive use of the NIR information. Although CBF recovers the textural details in the region, it still suffers from noise interference. Conversely, the proposed method not only balances the pixel intensity between two input images but also ensures all the detail information retaining in the fusion result. The quantitative results on NIR dataset are reported in Tab. V. We can easily notice that our method has the largest average values on four evaluation metrics except for VIF and $Q^{AB/F}$. These evaluation metrics including MI, CC, SSIM, and SCD indicate that our method has the strongest correlation with the source images. In contrast, GTF pays much attention to the influence of near-infrared images, and hence, it often has the highest average value in VIF and $Q^{AB/F}$ when the quality of input infrared images are high.

G. Limitations

As our network is trained according to the edge attention mechanism, the intermediate features pay more attention to texture details when reconstructs the images, which leads to our method might be less effective when source images contain significant noise. Fig. 15 shows the fusion results on different datasets, where the source images contain significant noise. It easy to notice that our method preserves most of the vital information from both sides, especially for the bush leaves in the second row. For ease of observation, regions in red boxes are highlighted below the images. Although our method is a lightweight network with less trained parameters, it remains a limitation of running time. In future work, we will consider a denoising module to integrate into our network with high efficiency.

H. Discussion on Un-Registered Image Fusion

Usually, multi-modal images will appear unregistered in many real scenes. To discuss the impact of unregistered images on our method, we initially synthesize mis-registered source

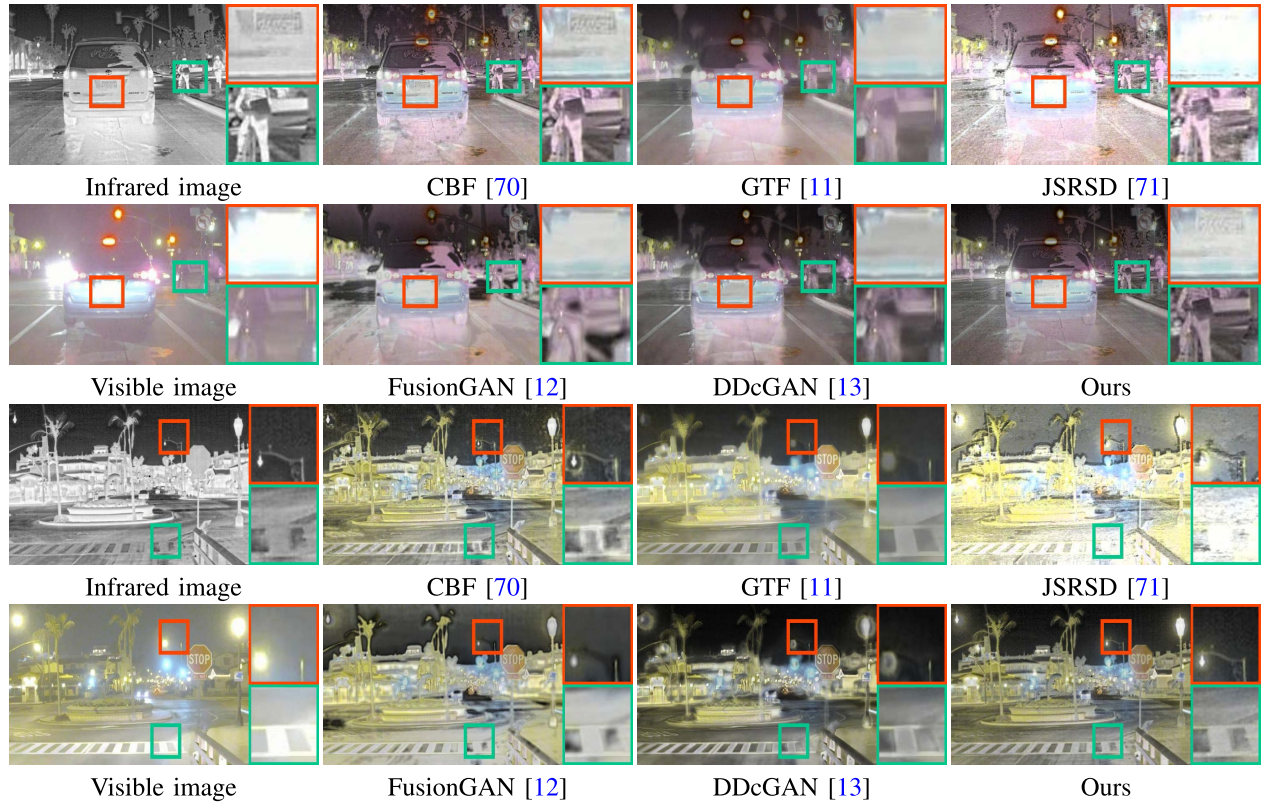


Fig. 13. Color visual results comparison between different methods on the FILR dataset. It can be seen clearly that only our method preserves the vehicle brand as shown in the red frame and figure target in the green frame.

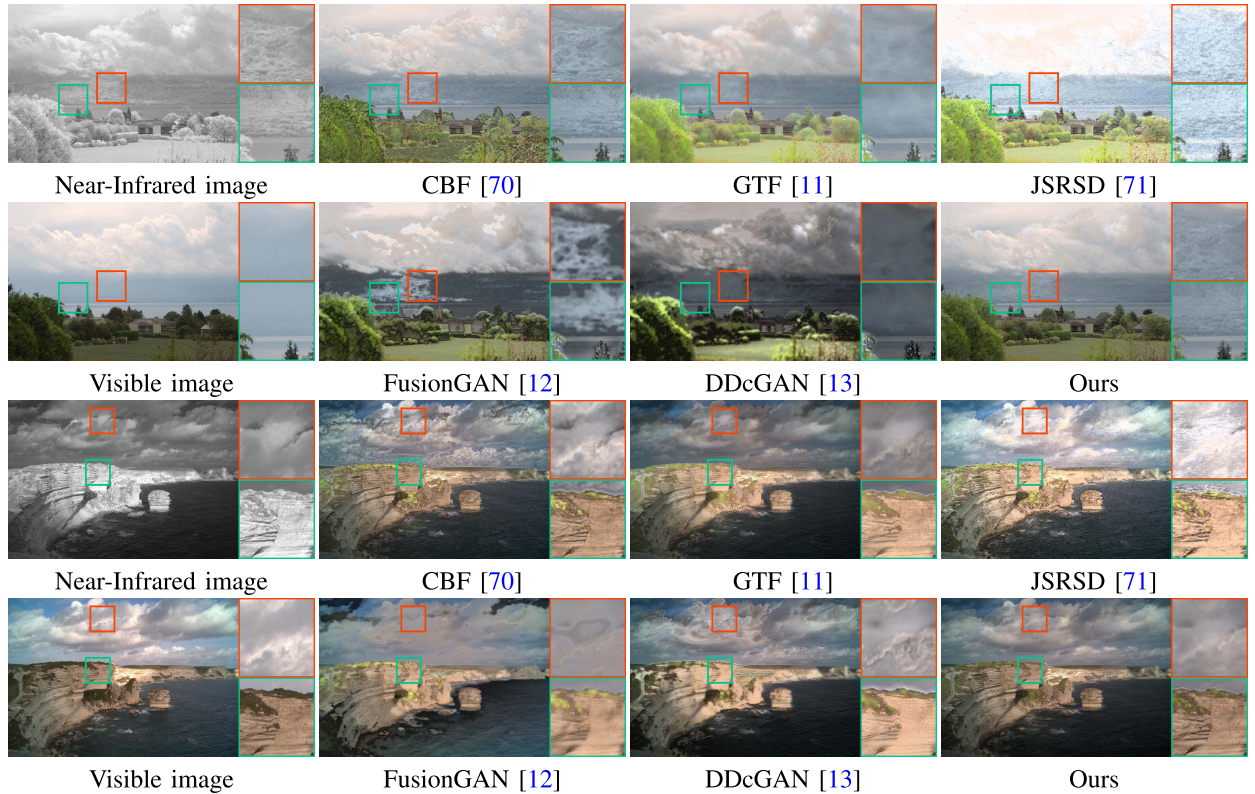


Fig. 14. Color visual results comparison between different methods on the NIR dataset. It is observed that our method presents appropriate color contrast without noise interference, and retain all the vital features in the fusion result.

images through performing random translations with $[-2,2]$, $[-5,5]$, and $[-10,10]$ pixels on the TNO datasets, respectively. Then we use the proposed method for fusing the mis-registered

source images. Visual results are shown in Fig.16, it can be seen that our method can deal with the slight unregistered images, which preserves a large part of vital information.

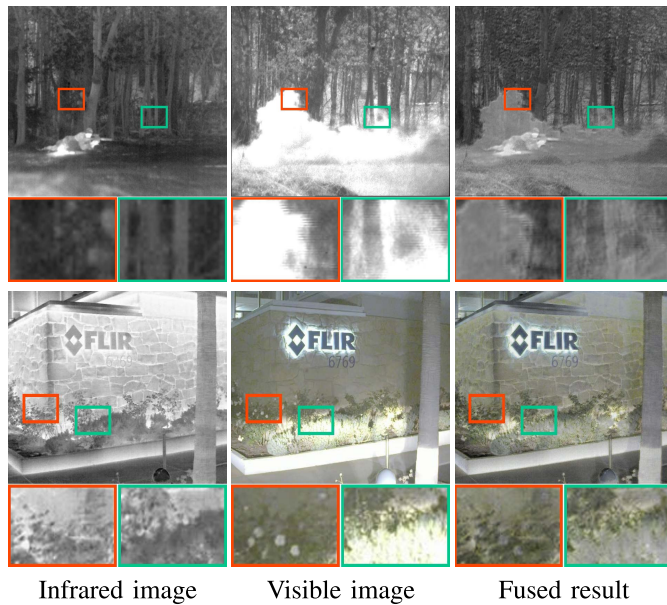


Fig. 15. Limitations of the proposed method. Our method is not effective in handling images with significant noise.

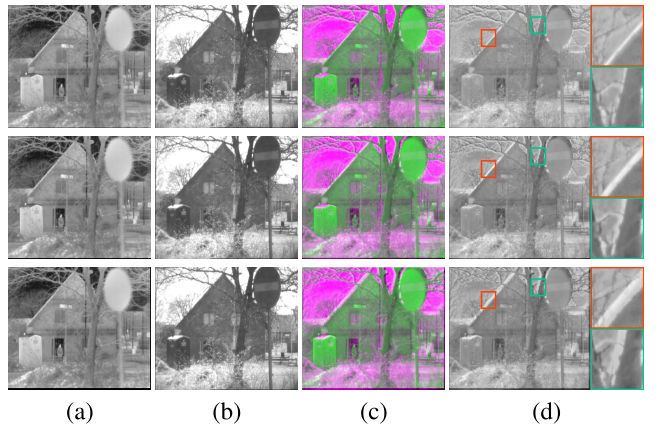


Fig. 16. Visual results of our fusion method on unregistered image pairs on the TNO dataset. We randomly offset the inputs, which can be seen as the black edge emerged in the images. From left to right: (a) infrared image, (b) visible image, (c) overlapping the source images before registered, (d) fused result. From top to bottom, the translations with $[-2,2]$, $[-5,5]$, and $[-10,10]$ pixels.

However, when pixel deviation is large, halos and artifacts emerge on the fusion result. The experimental results further show that our method has a certain tolerance for un-registered images.

V. CONCLUSION

Within this article, a novel deep learning method is proposed to efficiently solve the infrared and visible image fusion task. By means of the dilated convolution and the dense block, a coarse-to-fine feature extract module is proposed. This feature extractor can effectively refine deep salient features on the different scale without changing the image size. Besides, the edge attention mechanism is designed to fuse the multi-modal source images, which simultaneously avoids the degeneration of the details information in the fusion process and suppresses noise interference on the fused results. Sufficient experiments on the public datasets by comparing

nine state-of-the-art methods comprehensively illustrate that our final result not only represents an abundance of textural details but also gets rid of the noise and holes simultaneously. Meanwhile, extensive experiments on FLIR and NIR datasets verify the generalization and robustness of the proposed method. We also construct a new infrared and visible image fusion dataset, named RealStreet, to enrich the insufficient aligned infrared and visible image pairs. In our future work, we will apply our network to other multi-modal image fusion tasks, e.g., computed tomography (CT) and magnetic resonance imaging (MRI) image fusion, magnetic resonance imaging (MRI), and single-photon positron emission tomography (SPECT) image fusion.

REFERENCES

- [1] M. Yin, X. Liu, Y. Liu, and X. Chen, "Medical image fusion with parameter-adaptive pulse coupled neural network in nonsubsampling shearlet transform domain," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 1, pp. 49–64, Jan. 2019.
- [2] W. Zhao, D. Wang, and H. Lu, "Multi-focus image fusion with a natural enhancement via a joint multi-level deeply supervised convolutional neural network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 4, pp. 1102–1115, Apr. 2019.
- [3] R. Nie, J. Cao, D. Zhou, and W. Qian, "Multi-source information exchange encoding with PCNN for medical image fusion," *IEEE Trans. Circuits Syst. Video Technol.*, early access, May 29, 2020, doi: 10.1109/TCSVT.2020.2998696.
- [4] X. Zhang, P. Ye, and G. Xiao, "VIFB: A visible and infrared image fusion benchmark," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 104–105.
- [5] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object detection with discriminatively trained part-based models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010.
- [6] M. H. Wong *et al.*, "Nitroaromatic detection and infrared communication from wild-type plants using plant nanobionics," *Nature Mater.*, vol. 16, no. 2, pp. 264–271, 2017.
- [7] Q. Zhang, T. Xiao, N. Huang, D. Zhang, and J. Han, "Revisiting feature fusion for RGB-T salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Aug. 6, 2021, doi: 10.1109/TCSVT.2020.3014663.
- [8] X. Zhang, P. Ye, H. Leung, K. Gong, and G. Xiao, "Object fusion tracking based on visible and infrared images: A comprehensive review," *Inf. Fusion*, vol. 63, pp. 166–187, Nov. 2020.
- [9] Z. Duan, J. Lan, Y. Xu, B. Ni, L. Zhuang, and X. Yang, "Pedestrian detection via bi-directional multi-scale analysis," in *Proc. 25th ACM Int. Conf. Multimedia*, Oct. 2017, pp. 1023–1031.
- [10] J.-Y. Kwak, B. C. Ko, and J. Y. Nam, "Pedestrian tracking using online boosted random ferns learning in far-infrared imagery for safe driving at night," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 1, pp. 69–81, Jan. 2017.
- [11] J. Ma, C. Chen, C. Li, and J. Huang, "Infrared and visible image fusion via gradient transfer and total variation minimization," *Inf. Fusion*, vol. 31, pp. 100–109, Sep. 2016.
- [12] J. Ma, W. Yu, P. Liang, C. Li, and J. Jiang, "FusionGAN: A generative adversarial network for infrared and visible image fusion," *Inf. Fusion*, vol. 48, pp. 11–26, Aug. 2019.
- [13] H. Xu, P. Liang, W. Yu, J. Jiang, and J. Ma, "Learning a generative model for fusing infrared and visible images via conditional generative adversarial network with dual discriminators," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 3954–3960.
- [14] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
- [15] M. Pu, Y. Huang, Q. Guan, and Q. Zou, "GraphNet: Learning image pseudo annotations for weakly-supervised semantic segmentation," in *Proc. 26th ACM Int. Conf. Multimedia*, Oct. 2018, pp. 483–491.
- [16] R. Liu, Z. Jiang, X. Fan, and Z. Luo, "Knowledge-driven deep unrolling for robust image layer separation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1653–1666, May 2020.

- [17] S. Kuanar, C. Conly, and K. R. Rao, "Deep learning based HEVC in-loop filtering for decoder quality enhancement," in *Proc. Picture Coding Symp. (PCS)*, Jun. 2018, pp. 164–168.
- [18] S. Kuanar, V. Athitsos, D. Mahapatra, K. R. Rao, Z. Akhtar, and D. Dasgupta, "Low dose abdominal CT image reconstruction: An unsupervised learning based approach," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1351–1355.
- [19] R. Liu, P. Mu, J. Chen, X. Fan, and Z. Luo, "Investigating task-driven latent feasibility for nonconvex image modeling," *IEEE Trans. Image Process.*, vol. 29, pp. 7629–7640, 2020, doi: [10.1109/TIP.2020.3004733](https://doi.org/10.1109/TIP.2020.3004733).
- [20] J. Saeedi and K. Faez, "Infrared and visible image fusion using fuzzy logic and population-based optimization," *Appl. Soft Comput.*, vol. 12, no. 3, pp. 1041–1054, Mar. 2012.
- [21] Y.-S. Lee, Y.-M. Chan, L.-C. Fu, and P.-Y. Hsiao, "Near-Infrared-Based nighttime pedestrian detection using grouped part models," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 4, pp. 1929–1940, Aug. 2015.
- [22] H.-M. Hu, J. Wu, B. Li, Q. Guo, and J. Zheng, "An adaptive fusion algorithm for visible and infrared videos based on entropy and the cumulative distribution of gray levels," *IEEE Trans. Multimedia*, vol. 19, no. 12, pp. 2706–2719, Dec. 2017.
- [23] H. Guo, Y. Ma, X. Mei, and J. Ma, "Infrared and visible image fusion based on total variation and augmented lagrangian," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 34, no. 11, pp. 1961–1968, 2017.
- [24] W. Zhao, H. Lu, and D. Wang, "Multisensor image fusion and enhancement in spectral total variation domain," *IEEE Trans. Multimedia*, vol. 20, no. 4, pp. 866–879, Apr. 2018.
- [25] S. Li, X. Kang, and J. Hu, "Image fusion with guided filtering," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2864–2875, Jul. 2013.
- [26] Z. Zhu, H. Yin, Y. Chai, Y. Li, and G. Qi, "A novel multi-modality image fusion method based on image decomposition and sparse representation," *Inf. Sci.*, vol. 432, pp. 516–529, Mar. 2018.
- [27] G. Cui, H. Feng, Z. Xu, Q. Li, and Y. Chen, "Detail preserved fusion of visible and infrared images using regional saliency extraction and multi-scale image decomposition," *Opt. Commun.*, vol. 341, pp. 199–209, Apr. 2015.
- [28] Q. Zhang, Y. Liu, R. S. Blum, J. Han, and D. Tao, "Sparse representation based multi-sensor image fusion for multi-focus and multi-modality images: A review," *Inf. Fusion*, vol. 40, pp. 57–75, Mar. 2018.
- [29] X. Lu, B. Zhang, Y. Zhao, H. Liu, and H. Pei, "The infrared and visible image fusion algorithm based on target separation and sparse representation," *Infr. Phys. Technol.*, vol. 67, pp. 397–407, Nov. 2014.
- [30] R. Liu, J. Liu, Z. Jiang, X. Fan, and Z. Luo, "A bilevel integrated model with data-driven layer ensemble for multi-modality image fusion," *IEEE Trans. Image Process.*, vol. 30, pp. 1261–1274, 2021, doi: [10.1109/TIP.2020.3043125](https://doi.org/10.1109/TIP.2020.3043125).
- [31] Z. Zhao, S. Xu, C. Zhang, J. Liu, and J. Zhang, "Bayesian fusion for infrared and visible images," *Signal Process.*, vol. 177, Dec. 2020, Art. no. 107734.
- [32] V. N. Gangapure, S. Nanda, and A. S. Chowdhury, "Superpixel-based causal multisensor video fusion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 6, pp. 1263–1272, Jun. 2018.
- [33] G. Bhatnagar, Q. M. J. Wu, and Z. Liu, "Directive contrast based multimodal medical image fusion in NSCT domain," *IEEE Trans. Multimedia*, vol. 15, no. 5, pp. 1014–1024, Aug. 2013.
- [34] J. J. Lewis, R. J. O'Callaghan, S. G. Nikolov, D. R. Bull, and N. Canagarajah, "Pixel- and region-based image fusion with complex wavelets," *Inf. Fusion*, vol. 8, no. 2, pp. 119–130, Apr. 2007.
- [35] H. Li and X.-J. Wu, "DenseFuse: A fusion approach to infrared and visible images," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2614–2623, May 2019.
- [36] H. Li, X.-J. Wu, and J. Kittler, "Infrared and visible image fusion using a deep learning framework," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 2705–2710.
- [37] Z. Zhao, S. Xu, C. Zhang, J. Liu, P. Li, and J. Zhang, "DIDFuse: Deep image decomposition for infrared and visible image fusion," 2020, *arXiv:2003.09210*. [Online]. Available: <http://arxiv.org/abs/2003.09210>
- [38] F. Lahoud and S. Süsstrunk, "Fast and efficient zero-learning image fusion," 2019, *arXiv:1905.03590*. [Online]. Available: <http://arxiv.org/abs/1905.03590>
- [39] Y. Liu, X. Chen, H. Peng, and Z. Wang, "Multi-focus image fusion with a deep convolutional neural network," *Inf. Fusion*, vol. 36, pp. 191–207, Jul. 2017.
- [40] Q. Wu *et al.*, "Blind image quality assessment based on multichannel feature fusion and label transfer," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 26, no. 3, pp. 425–440, Mar. 2016.
- [41] K. Song, H. Yang, and Z. Yin, "Multi-scale attention deep neural network for fast accurate object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 10, pp. 2972–2985, Oct. 2019.
- [42] J. M. Alvarez, Y. LeCun, T. Gevers, and A. M. Lopez, "Semantic road segmentation via multi-scale ensembles of learned features," in *Proc. ECCV*. Berlin, Germany: Springer, 2012, pp. 586–595.
- [43] A. L. Da Cunha, J. Zhou, and M. N. Do, "The nonsubsampling contourlet transform: Theory, design, and applications," *IEEE Trans. Image Process.*, vol. 15, no. 10, pp. 3089–3101, Oct. 2006.
- [44] J. Ma, Z. Zhou, B. Wang, and H. Zong, "Infrared and visible image fusion based on visual saliency map and weighted least square optimization," *Infr. Phys. Technol.*, vol. 82, pp. 8–17, May 2017.
- [45] J. Ho Jang, Y. Bae, and J. Beom Ra, "Contrast-enhanced fusion of multisensor images using subband-decomposed multiscale retinex," *IEEE Trans. Image Process.*, vol. 21, no. 8, pp. 3479–3490, Aug. 2012.
- [46] X. Yan, H. Qin, J. Li, H. Zhou, and J.-G. Zong, "Infrared and visible image fusion with spectral graph wavelet transform," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 32, no. 9, pp. 1643–1652, 2015.
- [47] J. Liang, Y. He, D. Liu, and X. Zeng, "Image fusion using higher order singular value decomposition," *IEEE Trans. Image Process.*, vol. 21, no. 5, pp. 2898–2909, May 2012.
- [48] M. Kim, D. K. Han, and H. Ko, "Joint patch clustering-based dictionary learning for multimodal image fusion," *Inf. Fusion*, vol. 27, pp. 198–214, Jan. 2016.
- [49] T. Shibata, M. Tanaka, and M. Okutomi, "Versatile visible and near-infrared image fusion based on high visibility area selection," *J. Electron. Imag.*, vol. 25, no. 1, Jan. 2016, Art. no. 013016.
- [50] C. Yan *et al.*, "STAT: Spatial-temporal attention mechanism for video captioning," *IEEE Trans. Multimedia*, vol. 22, no. 1, pp. 229–241, Jan. 2020.
- [51] Y. Li, J. Zeng, S. Shan, and X. Chen, "Occlusion aware facial expression recognition using CNN with attention mechanism," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2439–2450, May 2019.
- [52] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
- [53] Z. Ji, K. Xiong, Y. Pang, and X. Li, "Video summarization with attention-based Encoder-Decoder networks," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1709–1717, Jun. 2020.
- [54] N. Liu, J. Han, and M.-H. Yang, "PiCANet: Learning pixel-wise contextual attention for saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3089–3098.
- [55] M. Suganuma, X. Liu, and T. Okatani, "Attention-based adaptive selection of operations for image restoration in the presence of unknown combined distortions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9039–9048.
- [56] R. Liu, S. Cheng, Y. He, X. Fan, Z. Lin, and Z. Luo, "On the convergence of learning-based iterative methods for nonconvex inverse problems," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 12, pp. 3027–3039, Dec. 2020.
- [57] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3640–3649.
- [58] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3156–3164.
- [59] Z. Zhang, H. Fu, H. Dai, J. Shen, Y. Pang, and L. Shao, "ET-net: A generic edge-attention guidance network for medical image segmentation," in *Proc. MICCAI*. Cham, Switzerland: Springer, 2019, pp. 442–450.
- [60] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8779–8788.
- [61] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," 2015, *arXiv:1511.07122*. [Online]. Available: <http://arxiv.org/abs/1511.07122>
- [62] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [63] J. Adler and S. Lunz, "Banach Wasserstein GAN," 2018, *arXiv:1806.06621*. [Online]. Available: <https://arxiv.org/abs/1806.06621>
- [64] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of wasserstein gans," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5767–5777.
- [65] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Proc. ECCV*. Cham, Switzerland: Springer, 2014, pp. 740–755.

- [66] G. Qu, D. Zhang, and P. Yan, "Information measure for performance of image fusion," *Electron. Lett.*, vol. 38, no. 7, pp. 313–315, Mar. 2002.
- [67] Y. Han, Y. Cai, Y. Cao, and X. Xu, "A new image fusion performance metric based on visual information fidelity," *Inf. Fusion*, vol. 14, no. 2, pp. 127–135, Apr. 2013.
- [68] C. S. Xydeas and V. Petrović, "Objective image fusion performance measure," *Electron. Lett.*, vol. 36, no. 4, pp. 308–309, 2000.
- [69] V. Aslantas and E. Bendes, "A new image quality metric for image fusion: The sum of the correlations of differences," *AEU-Int. J. Electron. Commun.*, vol. 69, no. 12, pp. 1890–1896, Dec. 2015.
- [70] B. K. Shreyamsha Kumar, "Image fusion based on pixel significance using cross bilateral filter," *Signal, Image Video Process.*, vol. 9, no. 5, pp. 1193–1204, Jul. 2015.
- [71] C. H. Liu, Y. Qi, and W. R. Ding, "Infrared and visible image fusion method based on saliency detection in sparse domain," *Infr. Phys. Technol.*, vol. 83, pp. 94–102, Jun. 2017.
- [72] Q. Du, H. Xu, Y. Ma, J. Huang, and F. Fan, "Fusing infrared and visible images of different resolutions via total variation model," *Sensors*, vol. 18, no. 11, p. 3827, Nov. 2018.
- [73] D. P. Bavarisetti, G. Xiao, and G. Liu, "Multi-sensor image fusion based on fourth order partial differential equations," in *Proc. 20th Int. Conf. Inf. Fusion*, Jul. 2017, pp. 1–9.
- [74] W. J. Roberts, J. A. A. Van, and F. Ahmed, "Assessment of image fusion procedures using entropy, image quality, and multispectral classification," *J. Appl. Remote Sens.*, vol. 2, no. 1, pp. 1–28, 2008.
- [75] M. Deshmukh and U. Bhosale, "Image fusion and image quality assessment of fused images," *Int. J. Image Process.*, vol. 4, no. 5, p. 484, 2010.
- [76] M. Brown and S. Susstrunk, "Multi-spectral SIFT for scene category recognition," in *Proc. CVPR*, Jun. 2011, pp. 177–184.



Jinyuan Liu received the M.S. degree in computer science from Dalian University, Dalian, China, in 2018. He is currently pursuing the Ph.D. degree in software engineering with the Dalian University of Technology, Dalian. He is also with the Key Laboratory for Ubiquitous Network and Service Software of Liaoning Province, Dalian University of Technology. His research interests include computer vision, image processing, and deep learning.



Xin Fan (Senior Member, IEEE) was born in 1977. He received the B.E. and Ph.D. degrees in information and communication engineering from Xi'an Jiaotong University, Xi'an, China, in 1998 and 2004, respectively.



He was with Oklahoma State University at Stillwater, Stillwater, OK, USA, from 2006 to 2007, as a Post-Doctoral Research Fellow. In 2009, he joined the School of Software, Dalian University of Technology, Dalian, China. He is currently a Professor with the DUT-RU International School of Information Science and Engineering, Dalian University of Technology. His current research interests include computational geometry and machine learning, and their applications to low-level image processing and diffusion tensor imaging magnetic resonance image analysis.



Ji Jiang is currently pursuing the B.S. degree in software engineering from the Dalian University of Technology, Dalian, China. He will pursue the master's degree in computer applications at Peking University, Shenzhen, China. He researches at the Key Laboratory for Ubiquitous Network and Service Software of Liaoning Province, Dalian University of Technology. His research interests include computer vision, image processing, and deep learning.



Risheng Liu (Member, IEEE) received the B.Sc. and Ph.D. degrees in mathematics from the Dalian University of Technology, Dalian, China, in 2007 and 2012, respectively.

He was a Visiting Scholar with the Robotic Institute, Carnegie Mellon University, from 2010 to 2012. He served as Hong Kong Scholar Research Fellow with The Hong Kong Polytechnic University from 2016 to 2017. He is currently a Professor with DUT-RU International School of Information Science and Engineering, Dalian University of Technology. He was awarded the "Outstanding Youth Science Foundation" of the National Natural Science Foundation of China. His research interests include machine learning, optimization, computer vision, and multimedia. He is a member of ACM. He was a co-recipient of the IEEE ICME Best Student Paper Award in 2014 and 2015. His two articles were also selected as Finalist of the Best Paper Award in ICME 2017.



Zhongxuan Luo received the B.S. and M.S. degrees from Jilin University, Changchun, China, in 1985 and 1988, respectively, and the Ph.D. degree from the Dalian University of Technology, Dalian, China, in 1991, all in computational mathematics.

He is currently a Full Professor with the School of Software Technology, Dalian University of Technology. His current research interests include computational geometry and computer vision.